

Wireless Communication Numerical Companion

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

Ch 3 · Radio Wave Propagation Numerical Problems & Solutions

This is the heaviest numerical chapter in the subject. A propagation numerical has appeared in every one of the 22 papers. Every problem below is a real PYQ, worked in full.

The formula sheet

- **Free space (Friis)** $P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}$, $\lambda = c/f$
- **Free space path loss** $PL = 20 \log \left(\frac{4\pi d}{\lambda} \right) = 32.44 + 20 \log f_{\text{MHz}} + 20 \log d_{\text{km}}$
- **Power flux density** $P_d = \frac{P_t G_t}{4\pi d^2} = \frac{|E|^2}{120\pi}$, **aperture** $A_e = \frac{G_r \lambda^2}{4\pi}$, $P_r = P_d A_e$
- **Matched receiver** $V_{\text{rms}} = \sqrt{P_r R_{\text{ant}}}$, open-circuit $V_{\text{ant}} = 2V_{\text{rms}}$
- **Two-ray** $P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4}$, $PL = 40 \log d - 20 \log h_t - 20 \log h_r$ (unity gains)
- **Log-distance** $PL(d) = PL(d_0) + 10n \log \left(\frac{d}{d_0} \right) + X_\sigma$
- **Okumura** $L_{50} = L_F + A_{mu} - G(h_{te}) - G(h_{re}) - G_{\text{AREA}}$, with $G(h_{te}) = 20 \log \frac{h_{te}}{200}$, $G(h_{re}) = 10 \log \frac{h_{re}}{3}$ for $h_{re} \leq 3$ m, $20 \log \frac{h_{re}}{3}$ for $3 < h_{re} < 10$ m
- **Hata (urban)** $L_{50} = 69.55 + 26.16 \log f_c - 13.82 \log h_{te} - a(h_{re}) + (44.9 - 6.55 \log h_{te}) \log d$
- **Delay spread** $\bar{\tau} = \frac{\sum P_k \tau_k}{\sum P_k}$, $\bar{\tau}^2 = \frac{\sum P_k \tau_k^2}{\sum P_k}$, $\sigma_\tau = \sqrt{\tau^2 - \bar{\tau}^2}$
- **Coherence** $B_c \approx \frac{1}{50\sigma_\tau}$ (90%), $B_c \approx \frac{1}{5\sigma_\tau}$ (50%), $f_m = \frac{v}{\lambda}$, $T_c \approx \frac{0.423}{f_m}$

Units discipline, the source of most lost marks: P in W for Friis, dBm only after conversion. d in metres in $20 \log(4\pi d/\lambda)$ but in km in the 32.44 form and in Hata. f in MHz in the 32.44 form and in Hata. Always write the unit beside every number.

Problem 1 Free space: received power, E-field, flux density, rms voltage

HOT 3/22 80 Ba, 81 Ba **8**, 78 Ch **2+2+2+2** · the classic four-part free space numerical

QUESTION AS PRINTED 2080 Baishakh · Q3 (b) · [8]

Assume free space propagation, a receiver is located 10 km away from a 50 w transmitter. The carrier frequency is 900 MHz. antenna gain at transmitter and receiver are 1 and 2 respectively, find:

- i) The power received at the receiver
- ii) The magnitude of E-field at the receiver antenna
- iii) The power flux density
- iv) The rms voltage applied to the receiver input.

The receiver antenna has a purely real impedance of 50 Ω and is matched to the receiver.

QUESTION AS PRINTED 2078 Chaitra · Q3 (b) · [2+2+2+2]

Assume free space propagation, a receiver is located 10 km away from a 50W transmitter. The carrier frequency is 900 MHz, antenna gain at transmitter and receiver is 1 and 2 respectively. Find: (i) The power received at the receiver. (ii) The magnitude of the E-field at the receiver. (iii) The rms voltage applied to the receiver input. (iv) The power flux density.

Same numbers, parts (iii) and (iv) swapped. "900 MHza" is the 80 Ba paper's typo.

Step 0: wavelength

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{900 \times 10^6} = 0.3333 \text{ m}$$

(i) Received power

- Friis, with $L = 1$ (no system loss stated):

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2} = \frac{50 \times 1 \times 2 \times (0.3333)^2}{(4\pi)^2 (10^4)^2}$$

$$= \frac{50 \times 2 \times 0.11111}{157.91 \times 10^8} = \frac{11.111}{1.5791 \times 10^{10}} = 7.0362 \times 10^{-10} \text{ W}$$

- In dBm:

$$P_r = 10 \log_{10}(7.0362 \times 10^{-10} \times 10^3) = \boxed{-61.53 \text{ dBm}} = -91.53 \text{ dBW}$$

(iii) Power flux density (do this before the E-field, it feeds it)

$$P_d = \frac{P_t G_t}{4\pi d^2} = \frac{50 \times 1}{4\pi (10^4)^2} = \frac{50}{1.2566 \times 10^9} = \boxed{3.9789 \times 10^{-8} \text{ W/m}^2}$$

(ii) Magnitude of the E-field

- Free space intrinsic impedance $\eta = 120\pi = 377 \Omega$, and $P_d = |E|^2/\eta$:

$$|E| = \sqrt{P_d \times 377} = \sqrt{3.9789 \times 10^{-8} \times 377} = \sqrt{1.5 \times 10^{-5}}$$

$$= \boxed{3.873 \times 10^{-3} \text{ V/m}} = 3.873 \text{ mV/m}$$

Cross-check: P_r from P_d and the aperture

$$A_e = \frac{G_r \lambda^2}{4\pi} = \frac{2 \times 0.11111}{12.566} = 0.017684 \text{ m}^2$$

$$P_r = P_d A_e = 3.9789 \times 10^{-8} \times 0.017684 = 7.036 \times 10^{-10} \text{ W} \quad \checkmark \text{ matches (i)}$$

(iv) rms voltage at the receiver input

- The antenna is **matched** to a purely real 50Ω load, so the power delivered is

$$P_r = \frac{V_{\text{rms}}^2}{R_{\text{ant}}} \Rightarrow V_{\text{rms}} = \sqrt{P_r R_{\text{ant}}}$$

$$= \sqrt{7.0362 \times 10^{-10} \times 50} = \sqrt{3.5181 \times 10^{-8}}$$

$$= \boxed{1.8757 \times 10^{-4} \text{ V}} = 187.57 \mu\text{V}$$

- If the question asks for the open-circuit antenna voltage V_{ant} instead, the matched load halves it, so

$$V_{\text{ant}} = 2V_{\text{rms}} = 375.13 \mu\text{V}$$

Read the wording: “applied to the receiver input” is V_{rms} ; “induced in the antenna” is V_{ant} .

Problem 2 Free space link with dBi antennas

PIN 1/22 80 Bh 6

QUESTION AS PRINTED 2080 Bhadra · Q2 (a) · [6]

A wireless communication transmitter has an output power of 165 watts at a carrier frequency of 325 MHz. It is connected to an antenna with a gain of 12 dBi. The receiving antenna is 15 km away and has a gain of 6 dBi. Calculate the power delivered to the receiver, considering free-space propagation. (Assume that there are no other losses or mismatches in the system.)

- **Gains are in dB, so work the whole problem in dB.** Converting them to linear and using Friis directly also works but is slower and more error-prone.
- **Step 1: transmit power in dBm.**

$$P_t = 10 \log_{10}(165 \times 10^3) = 10 \log_{10}(1.65 \times 10^5) = 52.17 \text{ dBm}$$

- **Step 2: free space path loss.** Using the km / MHz form:

$$PL = 32.44 + 20 \log_{10}(325) + 20 \log_{10}(15) \\ = 32.44 + 50.24 + 23.52 = 106.20 \text{ dB}$$

Check with $\lambda = 3 \times 10^8 / 325 \times 10^6 = 0.9231 \text{ m}$: $20 \log(4\pi \times 15000 / 0.9231) = 106.20 \text{ dB} \checkmark$

- **Step 3: link equation.**

$$P_r = P_t + G_t + G_r - PL = 52.17 + 12 + 6 - 106.20 \\ = \boxed{-36.03 \text{ dBm}}$$

- **Step 4: convert back to watts if asked.**

$$P_r = 10^{-36.03/10} \text{ mW} = 2.4966 \times 10^{-4} \text{ mW} = \boxed{249.7 \text{ nW}}$$

Problem 3 Full link budget with feeder and mismatch loss

PIN 1/22 72 Ash 8

QUESTION AS PRINTED 2072 Ashwin · Q3 · [8]

A BS transmitter has a power output of 10 watts operating at a frequency of 250 MHz. The transmitter is connected by 20 m of an RF coaxial cable, which has a loss of 3-dB/100 m specification, to an antenna that has a gain of 9 dBi. The receiving antenna is 25 km away and has a gain of 4 dBi. There is negligible loss in the receiver feeder line, but the receiver is mismatched; the receiving antenna and feeder cable are designed for 50 ohm impedance. The receiver impedance loss due to mismatch is of about 0.2 dB. Calculate the power delivered to the receiver, assuming free-space propagation.

- **Step 1: transmit power in dBm.**

$$P_t = 10 \log_{10}(10 \times 10^3) = 40.00 \text{ dBm}$$

- **Step 2: transmit feeder loss.**

$$L_{\text{coax}} = 20 \text{ m} \times \frac{3 \text{ dB}}{100 \text{ m}} = 0.60 \text{ dB}$$

- **Step 3: free space path loss at 25 km, 250 MHz.**

$$PL = 32.44 + 20 \log_{10}(250) + 20 \log_{10}(25) \\ = 32.44 + 47.96 + 27.96 = 108.36 \text{ dB}$$

- **Step 4: assemble the budget.** Every gain adds, every loss subtracts.

Item	Value	Note
Transmitter output	+40.00 dBm	10 W
Transmit feeder (coax)	−0.60 dB	20 m at 3 dB/100 m
Transmit antenna gain	+9.00 dB	9 dBi
Free space path loss	−108.36 dB	25 km, 250 MHz
Receive antenna gain	+4.00 dB	4 dBi
Receive feeder loss	−0.00 dB	stated negligible
Receiver mismatch loss	−0.20 dB	50 Ω system
Power at receiver	−56.16 dBm	

$$P_r = 40.00 - 0.60 + 9.00 - 108.36 + 4.00 - 0.20 = \boxed{-56.16 \text{ dBm}} \\ = 10^{-56.16/10} \text{ mW} = 2.42 \times 10^{-6} \text{ mW} = \boxed{2.42 \text{ nW}}$$

Marking tip: draw the budget as a table like this one. Examiners give method marks for a laid-out budget even when one entry is wrong.

Problem 4 Radio coverage range from a receiver threshold

PIN 1/22 70 Bh 6

QUESTION AS PRINTED 2070 Bhadra · Q4 · [6]

Determine the radio coverage range of a base station that transmits a RF signal at 150 W, given the receiver threshold level is -104 dBm. Assume that the path loss at the first meter is 15 dB in a mobile radio propagation condition. (Path loss exponent = 4)

- **Step 1: transmit power in dBm.**

$$P_t = 10 \log_{10}(150 \times 10^3) = 51.76 \text{ dBm}$$

- **Step 2: maximum allowable path loss.** The link works as long as $P_r \geq -104$ dBm, so

$$PL_{\max} = P_t - P_{r,\min} = 51.76 - (-104) = 155.76 \text{ dB}$$

- **Step 3: log-distance model with $d_0 = 1$ m.**

$$PL(d) = PL(d_0) + 10n \log_{10}\left(\frac{d}{d_0}\right) = 15 + 40 \log_{10}(d)$$

- **Step 4: solve for d .**

$$\begin{aligned} 155.76 &= 15 + 40 \log_{10}(d) \\ \log_{10}(d) &= \frac{155.76 - 15}{40} = \frac{140.76}{40} = 3.519 \\ d &= 10^{3.519} = \boxed{3304 \text{ m} \approx 3.30 \text{ km}} \end{aligned}$$

- **Sanity check:** at 3.3 km with $n = 4$ the loss beyond 1 m is $40 \log(3304) = 140.8$ dB, plus the 15 dB first-metre loss gives 155.8 dB, which is exactly the budget. ✓

Problem 5 Monopole aperture and two-ray received power

PIN 1/22 71 Bh 6

QUESTION AS PRINTED 2071 Bhadra · Q4 · [6]

A mobile is located 5 km away from a base station and a vertical $\lambda/4$ monopole antenna with a gain of 2.55 dB to receive cellular radio signals. The electric field at 1 km from the transmitter is measured to be 10^{-3} V/m. The carrier frequency used for this system is 900 MHz.

- Find the length and effective aperture of the receiving antenna.
- Find the received power at the mobile using two ray ground reflection model assuming the height of the transmitting antenna is 50 m and the receiving antenna is 1.5 m above ground.

(a) Length and effective aperture

–

$$\lambda = \frac{3 \times 10^8}{900 \times 10^6} = 0.3333 \text{ m}$$

- **Length of a quarter-wave monopole:**

$$L = \frac{\lambda}{4} = \frac{0.3333}{4} = \boxed{0.0833 \text{ m} = 8.33 \text{ cm}}$$

- **Gain in linear terms:**

$$G_r = 10^{2.55/10} = 1.7989$$

- **Effective aperture:**

$$A_e = \frac{G_r \lambda^2}{4\pi} = \frac{1.7989 \times 0.11111}{12.566} = \boxed{0.0159 \text{ m}^2}$$

(b) Received power from the two-ray model

- **Step 1: the two-ray E-field.** For d well beyond the crossover distance,

$$|E_{\text{tot}}(d)| = \frac{2E_0 d_0}{d} \cdot \frac{2\pi h_t h_r}{\lambda d}$$

with the reference $E_0 d_0 = 10^{-3} \times 1000 = 1 \text{ V}$ (measured at $d_0 = 1 \text{ km}$).

– **Step 2: substitute.**

$$\begin{aligned}
 |E_{\text{tot}}| &= \frac{2 \times 1}{5000} \cdot \frac{2\pi \times 50 \times 1.5}{0.3333 \times 5000} \\
 &= 4 \times 10^{-4} \times \frac{471.24}{1666.7} = 4 \times 10^{-4} \times 0.28274 \\
 &= \boxed{1.131 \times 10^{-4} \text{ V/m}}
 \end{aligned}$$

– **Step 3: convert field to power through the aperture.**

$$\begin{aligned}
 P_r &= \frac{|E_{\text{tot}}|^2}{120\pi} \cdot A_e = \frac{(1.131 \times 10^{-4})^2}{377} \times 0.0159 \\
 &= \frac{1.2791 \times 10^{-8}}{377} \times 0.0159 = 3.3928 \times 10^{-11} \times 0.0159 \\
 &= \boxed{5.397 \times 10^{-13} \text{ W} = -92.68 \text{ dBm}}
 \end{aligned}$$

- **Check the model is valid here:** crossover distance $d_c = 4\pi h_t h_r / \lambda = 4\pi(50)(1.5)/0.3333 = 2827 \text{ m}$. Since $d = 5000 \text{ m} > d_c$, the $1/d^2$ field approximation holds. ✓

Problem 6 Free space path loss of a cordless phone

HOT 2/22 77 Ch 6, 80 Ch 2+4

QUESTION AS PRINTED 2077 Chaitra · Q3 · [6]

Given a cordless phone operating at 52000 kHz frequency which has a range of 50m. Determine the free space path loss.

QUESTION AS PRINTED 2080 Chaitra · Q4 (b) · [2+4]

Define Doppler's effect in Small Scale Fading. (i) Given a cordless phone operating at 52000 kHz frequency which has a range of 50m. Determine the free space path loss.

– **Step 1: sort out the units.**

$$f = 52000 \text{ kHz} = 52 \text{ MHz} = 52 \times 10^6 \text{ Hz}$$

The “52000 kHz” is deliberately awkward. Convert it first.

– **Step 2: wavelength.**

$$\lambda = \frac{3 \times 10^8}{52 \times 10^6} = 5.769 \text{ m}$$

– **Step 3: path loss.**

$$\begin{aligned}
 PL &= 20 \log_{10} \left(\frac{4\pi d}{\lambda} \right) = 20 \log_{10} \left(\frac{4\pi \times 50}{5.769} \right) \\
 &= 20 \log_{10}(108.9) = \boxed{40.74 \text{ dB}}
 \end{aligned}$$

– **Cross-check with the km / MHz form, $d = 0.05 \text{ km}$:**

$$PL = 32.44 + 20 \log(52) + 20 \log(0.05) = 32.44 + 34.32 - 26.02 = 40.74 \text{ dB} \quad \checkmark$$

- **Note how low this is** compared with the 100 dB-plus figures elsewhere: short range and a long wavelength both help. That is why cordless phones need only milliwatts.

Problem 7 dBm, dBW and received power at two distances

PIN 1/22 81 Ba 1+1+2+2

QUESTION AS PRINTED 2081 Baishakh · Q3 (b) · [1+1+2+2]

If a transmitter produces 50 watts of power, express the transmit power in units of (i) dBm, and (ii) dBW. If 50 watts is applied to a unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100 m from the antenna. What is P_r (10 km)? Assume unity gain for the receiver antenna.

– (i)

$$P_t [\text{dBm}] = 10 \log_{10}(50 \times 10^3) = 10 \log_{10}(5 \times 10^4) = \boxed{46.99 \text{ dBm}}$$

– (ii)

$$P_t [\text{dBW}] = 10 \log_{10}(50) = \boxed{16.99 \text{ dBW}}$$

Always dBm = dBW + 30. ✓

– (iii) At $d = 100 \text{ m}$, $\lambda = 0.3333 \text{ m}$, $G_t = G_r = 1$:

$$P_r = \frac{50 \times 1 \times 1 \times (0.3333)^2}{(4\pi)^2 (100)^2} = \frac{5.5556}{157.91 \times 10^4} = 3.5181 \times 10^{-6} \text{ W}$$

$$= 10 \log_{10}(3.5181 \times 10^{-3} \text{ mW}) = \boxed{-24.54 \text{ dBm}}$$

– (iv) At $d = 10 \text{ km}$, use the $1/d^2$ scaling instead of recomputing:

$$P_r(10 \text{ km}) = P_r(100 \text{ m}) \left(\frac{100}{10000} \right)^2 = 3.5181 \times 10^{-6} \times 10^{-4}$$

$$= 3.5181 \times 10^{-10} \text{ W} = \boxed{-64.54 \text{ dBm}}$$

– **The shortcut in dB:** the distance grew by a factor 100, ie 2 decades, and free space costs **20 dB per decade**, so $-24.54 - 40 = -64.54 \text{ dBm}$. ✓

Problem 8 Far-field (Fraunhofer) distance

PIN 1/22 79 Ch **3+3**

QUESTION AS PRINTED 2079 Chaitra · Q4 (b) · [3+3]

Explain basic propagation models. Compute the far field distance for an antenna with maximum dimension of 1 m and operating frequency of 900 MHz.

–

$$\lambda = \frac{3 \times 10^8}{900 \times 10^6} = 0.3333 \text{ m}$$

–

$$d_f = \frac{2D^2}{\lambda} = \frac{2 \times (1)^2}{0.3333} = \boxed{6 \text{ m}}$$

– **Verify both side conditions:**

– $d_f \gg D$: $6 \gg 1$ ✓

– $d_f \gg \lambda$: $6 \gg 0.3333$ ✓

– **Meaning:** the Friis equation may only be used beyond 6 m from this antenna. Closer in, the wavefront is not yet planar and the formula is meaningless.

Problem 9 Okumura model: median path loss and received power

HOT 4/22 74 Ma **4+4**, 80 Bh **4+2+2**, 82 Bh **5** · **the single most repeated numerical in the chapter**

QUESTION AS PRINTED 2080 Bhadra · Q3 (b) · [4+2+2]

Determine the mean path loss using Okumura's model for $d = 50 \text{ km}$, $h_{te} = 100 \text{ m}$, $h_{re} = 10 \text{ m}$ in a suburban environment. If the base station transmitter radiates an EIRP of 1kW at a carrier frequency of 900 MHz. Find EIRP (dBm) and the power at the receiver where the gain at receiving antenna is 10 dB. The medium attenuation factor is 43 dB and the gain due to area is 9 dB for a 50 km distance and 900 MHz frequency of operation.

QUESTION AS PRINTED 2082 Bhadra · Q3 (b) · [5]

Find the mean path loss using Okumura's Model for $d = 50$ km, $h_{te} = 100$ m, $h_{re} = 10$ m in a suburban environment. If the BS transmitter radiates an EIRP of 1 kW at a carrier frequency of 900 MHz, find EIRP in dBm and the power at the receiver where gain of receiving antenna is 10 dB. (Given: $A_{mu}(f, d) = 43$ dB, $G_{Area}(f) = 9$ dB)

QUESTION AS PRINTED 2074 Magh · Q3 · [4+4]

Explain Okumura model of outdoor radio propagation. Determine the median path loss for T-R separation of 50 km, transmit antenna effective height of 100 m and receive antenna effective height of 10 m in a suburban area with correction factor of 9 dB at 900 MHz. Assume median attenuation relative to free space 43 dB.

One set of numbers, three papers. "EIRP of 1 kW" is the 82 Bh paper's own typo.

Step 1: free space path loss at 50 km, 900 MHz

$$\lambda = 0.3333 \text{ m}$$

$$L_F = 20 \log_{10} \left(\frac{4\pi \times 50 \times 10^3}{0.3333} \right) = 20 \log_{10}(1.885 \times 10^6) = 125.51 \text{ dB}$$

Check: $32.44 + 20 \log(900) + 20 \log(50) = 32.44 + 59.08 + 33.98 = 125.50 \text{ dB} \checkmark$

Step 2: antenna height gain factors

- Base station, $h_{te} = 100$ m (valid, since $30 < 100 < 1000$ m):

$$G(h_{te}) = 20 \log_{10} \left(\frac{100}{200} \right) = 20 \log_{10}(0.5) = -6.02 \text{ dB}$$

Negative, because 100 m is below the 200 m reference.

- Mobile, $h_{re} = 10$ m. Since $3 < h_{re} < 10$ m the **20 log** branch applies:

$$G(h_{re}) = 20 \log_{10} \left(\frac{10}{3} \right) = 20 \log_{10}(3.333) = +10.46 \text{ dB}$$

Choosing the wrong branch here is the standard error. Check h_{re} against 3 m first.

Step 3: assemble the Okumura result

$$\begin{aligned} L_{50} &= L_F + A_{mu} - G(h_{te}) - G(h_{re}) - G_{AREA} \\ &= 125.51 + 43 - (-6.02) - (10.46) - 9 \\ &= 125.51 + 43 + 6.02 - 10.46 - 9 = \boxed{155.07 \text{ dB}} \end{aligned}$$

Step 4 (80 Bh, 82 Bh): EIRP and received power

$$\text{EIRP} = 1 \text{ kW} = 1000 \text{ W} \Rightarrow 10 \log_{10}(1000 \times 10^3) = \boxed{60 \text{ dBm}}$$

$$P_r = \text{EIRP} - L_{50} + G_r = 60 - 155.07 + 10 = \boxed{-85.07 \text{ dBm}}$$

$$= 10^{-85.07/10} \text{ mW} = 3.11 \times 10^{-9} \text{ mW} = 3.11 \text{ pW}$$

- Interpretation:** -85 dBm is a usable but not comfortable level. Typical GSM sensitivity is around -104 dBm , so this link has about 19 dB of margin.

Problem 10 Okumura run backwards: find the T-R separation

QUESTION AS PRINTED 2077 Chaitra · Q4 (b) · [8]

Employing the Okumura model compute the transmitter and receiver separation distance if median loss is 167 dB when the carrier frequency is 2.1 GHz. Assume height of transmitting antenna is 40 m, height of receiving antenna is 2 m, for a large city. [$A_{mu} = 34$ dB, $G_{area} = 0$ dB]

- **Step 1: the height gain factors.**

$$G(h_{te}) = 20 \log_{10} \left(\frac{40}{200} \right) = 20 \log_{10}(0.2) = -13.98 \text{ dB}$$

$$h_{re} = 2 \text{ m} \leq 3 \text{ m} \Rightarrow G(h_{re}) = 10 \log_{10} \left(\frac{2}{3} \right) = -1.76 \text{ dB}$$

Here the 10 log branch applies. Compare with Problem 9, where it was 20 log.

- **Step 2: rearrange Okumura for L_F .**

$$L_{50} = L_F + A_{mu} - G(h_{te}) - G(h_{re}) - G_{area}$$

$$L_F = L_{50} - A_{mu} + G(h_{te}) + G(h_{re}) + G_{area}$$

$$= 167 - 34 + (-13.98) + (-1.76) + 0 = \boxed{117.26 \text{ dB}}$$

- **Step 3: invert the free space formula.**

$$\lambda = \frac{3 \times 10^8}{2.1 \times 10^9} = 0.1429 \text{ m}$$

$$L_F = 20 \log_{10} \left(\frac{4\pi d}{\lambda} \right) \Rightarrow d = \frac{\lambda}{4\pi} 10^{L_F/20}$$

$$= \frac{0.1429}{12.566} \times 10^{117.26/20} = 0.011370 \times 10^{5.863}$$

$$= 0.011370 \times 7.2934 \times 10^5 = \boxed{8292 \text{ m} \approx 8.29 \text{ km}}$$

- **Sanity check:** substituting $d = 8.29$ km back gives $L_F = 32.44 + 20 \log(2100) + 20 \log(8.292) = 32.44 + 66.44 + 18.38 = 117.26 \text{ dB} \checkmark$

Problem 11 Hata model, and comparison with free space

PIN 1/22 79 Bh 4+4

QUESTION AS PRINTED 2079 Bhadra · Q3 (b) · [4+4]

In mobile propagation in a cellular system, find the correction factor and pathloss for a medium size city assuming carrier frequency as 950 MHz, height of transmitting antenna at base station is 45 m, propagation distance between antennas is 10 km and height of receiving antenna in mobile station is 5 m. Compute free space pathloss and compare it with Hata pathloss.

- **Step 1: mobile antenna correction factor, small/medium city branch:**

$$a(h_{re}) = (1.1 \log_{10} f_c - 0.7)h_{re} - (1.56 \log_{10} f_c - 0.8)$$

$$\log_{10}(950) = 2.9777$$

$$= (1.1 \times 2.9777 - 0.7)(5) - (1.56 \times 2.9777 - 0.8)$$

$$= (3.2755 - 0.7)(5) - (4.6452 - 0.8)$$

$$= 2.5755 \times 5 - 3.8452 = 12.8775 - 3.8452 = \boxed{9.032 \text{ dB}}$$

- **Step 2: the four Hata terms.** $\log_{10}(45) = 1.6532$, $\log_{10}(10) = 1$.

Term	Evaluation	Value (dB)
Constant	69.55	+69.550
$26.16 \log f_c$	26.16×2.9777	+77.897
$-13.82 \log h_{te}$	-13.82×1.6532	-22.847
$-a(h_{re})$	-9.032	-9.032
$(44.9 - 6.55 \log h_{te}) \log d$	$(44.9 - 10.828) \times 1$	+34.072
L_{50} (urban)		+149.64

-

$$L_{50}(\text{urban}) = 149.64 \text{ dB}$$

- **Step 3: free space path loss at the same distance.**

$$PL_{FS} = 32.44 + 20 \log(950) + 20 \log(10) = 32.44 + 59.56 + 20 = 112.00 \text{ dB}$$

- **Step 4: compare.**

$$L_{50} - PL_{FS} = 149.64 - 112.00 = 37.64 \text{ dB}$$

- **Interpretation:** the urban clutter costs almost **38 dB** over free space, ie a factor of about 5800 in power. This gap is exactly why free space is useless for cell planning and empirical models exist.
- **Validity check:** $f_c = 950 \text{ MHz}$ is inside 150–1500 MHz ✓, $d = 10 \text{ km}$ inside 1–20 km ✓, $h_{te} = 45 \text{ m}$ inside 30–200 m ✓, $h_{re} = 5 \text{ m}$ inside 1–10 m ✓. **State this; it carries a mark.**

Problem 12 Hata run backwards: maximum cell radius and area

PIN 1/22 73 Ma 6

QUESTION AS PRINTED 2073 Magh · Q3 (b) · [6]

Let us consider a medium sized city and assume the typical GSM downlink parameters. The Base Station (BS) is transmitting with power 50W. The minimum acceptable received power at Mobile Station (MS) is -91 dBm . The carrier frequency is 900 MHz, the height of BS is 30m and height of MS is 1m. Estimate the maximum cell radius and corresponding cell area using Hata Model.

The paper prints the full Hata model box in its header: $L_{50}(\text{urban}) = 69.55 + 26.16 \log f_c - 13.82 \log h_{te} - a(h_{re}) + (44.9 - 6.55 \log h_{te}) \log d$, with $a(h_{re}) = (1.1 \log f_c - 0.7)h_{re} - (1.56 \log f_c - 0.8) \text{ dB}$ for a medium sized city.

- **Step 1: allowable path loss.**

$$P_t = 10 \log_{10}(50 \times 10^3) = 46.99 \text{ dBm}$$

$$PL_{\max} = 46.99 - (-91) = 137.99 \text{ dB}$$

- **Step 2: correction factor, medium city, $\log_{10}(900) = 2.9542$:**

$$a(h_{re}) = (1.1 \times 2.9542 - 0.7)(1) - (1.56 \times 2.9542 - 0.8)$$

$$= (3.2496 - 0.7) - (4.6086 - 0.8) = 2.5496 - 3.8086 = -1.259 \text{ dB}$$

A negative correction is normal for a 1 m mobile antenna.

- **Step 3: split Hata into $L_{50} = A + B \log d$. With $\log_{10}(30) = 1.4771$:**

$$A = 69.55 + 26.16(2.9542) - 13.82(1.4771) - (-1.259)$$

$$= 69.55 + 77.281 - 20.413 + 1.259 = 127.678$$

$$B = 44.9 - 6.55(1.4771) = 44.9 - 9.675 = 35.225$$

- **Step 4: solve for d .**

$$137.99 = 127.678 + 35.225 \log_{10} d$$

$$\log_{10} d = \frac{137.99 - 127.678}{35.225} = \frac{10.312}{35.225} = 0.29274$$

$$d = 10^{0.29274} = 1.962 \text{ km}$$

- **Step 5: cell area.** Treating the cell as a regular hexagon of radius $R = d$:

$$A_{\text{cell}} = 2.598 R^2 = 2.598 \times (1.962)^2 = 2.598 \times 3.850 = 10.00 \text{ km}^2$$

If the cell is taken as a circle instead, $\pi R^2 = 12.10 \text{ km}^2$. State which you used.

- **Note $d = 1.96 \text{ km}$ is just below the Hata validity floor of 1 km**, so the answer sits at the edge of the model's range. Mention this: it is worth a mark.

Problem 13 Hata in a large city

PIN 1/22 73 Bh 7

QUESTION AS PRINTED 2073 Bhadra · Q3 (b) · [7]

Determine the propagation path loss for a radio signal 900 MHz cellular system operating in a large urban city, with a base station transmitter antenna height of 100 m and mobile receiver antenna height of 2m. The mobile unit is located at a distance of 4 km. Use the Hata propagation path loss model.

Hints printed with the question: $L_{50} = 69.55 + 26.16 \log f_c - 13.82 \log h_t - \alpha(h_r) + (44.9 - 6.55 \log h_t) \log d$; $\alpha(h_r) = (1.1 \log f_c - 0.7)h_r - (1.56 \log f_c - 0.8)$ for a small to medium city, $8.29(\log 1.54h_r)^2 - 1.1$ for a large city with $f_c \leq 300$ MHz, and $3.2(\log 11.75h_r)^2 - 4.97$ for a large city with $f_c > 300$ MHz.

- **Step 1: pick the correct $a(h_{re})$ branch.** Large city, $f_c = 900 > 300$ MHz, so

$$\begin{aligned} a(h_{re}) &= 3.2 (\log_{10}(11.75 h_{re}))^2 - 4.97 \\ &= 3.2 (\log_{10}(11.75 \times 2))^2 - 4.97 = 3.2 (\log_{10}(23.5))^2 - 4.97 \\ &= 3.2(1.3711)^2 - 4.97 = 3.2 \times 1.8799 - 4.97 = 6.0156 - 4.97 = \boxed{1.045 \text{ dB}} \end{aligned}$$

- **Step 2: the Hata terms.** $\log_{10}(900) = 2.9542$, $\log_{10}(100) = 2$, $\log_{10}(4) = 0.6021$.

Term	Evaluation	Value (dB)
Constant	69.55	+69.550
$26.16 \log f_c$	26.16×2.9542	+77.281
$-13.82 \log h_{te}$	-13.82×2	-27.640
$-a(h_{re})$	-1.045	-1.045
$(44.9 - 6.55 \log h_{te}) \log d$	$(44.9 - 13.10) \times 0.6021$	+19.147
L_{50} (urban)		+137.29

$$\boxed{L_{50} = 137.29 \text{ dB}}$$

Compare with Problem 11: there the mobile was 5 m high in a medium city and the loss was 149.6 dB at 10 km. Here the BS is twice as high and the distance is 2.5 times shorter, so 137.3 dB is consistent.

Problem 14 Two-ray Vs free space path loss

PIN 1/22 70 Ma 4+4

QUESTION AS PRINTED 2070 Magh · Q3 · [4+4]

Determine the propagation path loss for signal at 800 MHZ, with a transmitting antenna height of 30 m and a receiving antenna height of 2 m, over a distance of 10 km, using two-ray mobile point-to-point propagation model. How is it compared with that of free-space propagation path loss model?

- **Step 1: check the model is valid.**

$$\begin{aligned} \lambda &= \frac{3 \times 10^8}{800 \times 10^6} = 0.375 \text{ m} \\ d_c &= \frac{4\pi h_t h_r}{\lambda} = \frac{4\pi \times 30 \times 2}{0.375} = \frac{754.0}{0.375} = 2011 \text{ m} = 2.01 \text{ km} \end{aligned}$$

Since $d = 10 \text{ km} > d_c$, the d^{-4} form applies. ✓ **Always do this check first: below d_c the two-ray model oscillates and this formula is wrong.**

- **Step 2: two-ray path loss (unity antenna gains):**

$$\begin{aligned} PL &= 40 \log_{10} d - (20 \log_{10} h_t + 20 \log_{10} h_r) \\ &= 40 \log_{10}(10000) - 20 \log_{10}(30) - 20 \log_{10}(2) \\ &= 40(4) - 29.54 - 6.02 = 160 - 35.56 = \boxed{124.44 \text{ dB}} \end{aligned}$$

- **Step 3: free space path loss at the same distance.**

$$PL_{FS} = 32.44 + 20 \log(800) + 20 \log(10) = 32.44 + 58.06 + 20 = \boxed{110.50 \text{ dB}}$$

– **Step 4: comparison.**

$$PL_{2\text{-ray}} - PL_{FS} = 124.44 - 110.50 = \boxed{13.94 \text{ dB worse}}$$

- **Why the two-ray model predicts more loss:** the ground-reflected ray arrives almost exactly out of phase with the direct ray and **partially cancels** it. Free space ignores that ray entirely, so it is over-optimistic.
- **Note also:** the two-ray answer contains no λ term, so it is **frequency independent**, while the free space answer is not.

Problem 15 Link budget with Okumura: reverse-link distance

PIN 1/22 76 Bh 10

QUESTION AS PRINTED 2076 Bhadra · Q3 (b) · [10]

Estimate the appropriate distance that should be maintained for reverse link between one BTS and mobile with appropriate link budget diagram.

- Mobile is connected to antenna with 20dBi gain, with a transmitting power of 15 dBm and a receive sensitivity of -75dBm .
- BTS is connected to antenna with 5 dBi gain, with a transmitting power of 20dBm and a receive sensitivity of -80dBm .
- Cables in both systems are short, with a loss of 3-dB at each side at 900 MHz frequency of operation.

Use Okumura's model for mean path loss where $G(\text{Area}) = 9\text{dB}$, $A_{mu} = 43 \text{ dB}$, $h_{re} = 10\text{m}$ and $h_{te} = 100\text{m}$. (Link margin (reverse link) = 10 dB).

- **Step 1: identify the reverse link.** Reverse = **mobile transmits, BTS receives**. So use the **mobile's TX power** and the **BTS's sensitivity**.
- **Step 2: maximum allowable path loss on the reverse link.**

Item	Value	Note
Mobile TX power	+15 dBm	
Mobile antenna gain	+20 dB	20 dBi
Mobile cable loss	−3 dB	
BTS antenna gain	+5 dB	5 dBi
BTS cable loss	−3 dB	
BTS sensitivity	−(−80) dBm	the link must land at or above this
Link margin	−10 dB	fading reserve
PL_{max}	104 dB	

$$PL_{\max} = 15 + 20 - 3 + 5 - 3 + 80 - 10 = \boxed{104 \text{ dB}}$$

- **Step 3: convert this L_{50} into a free space loss.**

$$G(h_{te}) = 20 \log\left(\frac{100}{200}\right) = -6.02 \text{ dB}, \quad G(h_{re}) = 20 \log\left(\frac{10}{3}\right) = +10.46 \text{ dB}$$

$$\begin{aligned} L_F &= L_{50} - A_{mu} + G(h_{te}) + G(h_{re}) + G_{\text{AREA}} \\ &= 104 - 43 + (-6.02) + 10.46 + 9 = 74.44 \text{ dB} \end{aligned}$$

- **Step 4: solve for d .**

$$\begin{aligned} d &= \frac{\lambda}{4\pi} 10^{L_F/20} = \frac{0.3333}{12.566} \times 10^{74.44/20} = 0.026526 \times 5271 \\ &= \boxed{139.8 \text{ m} \approx 0.14 \text{ km}} \end{aligned}$$

- **Step 5: check the forward link too.** Forward = BTS transmits, mobile receives:

$$PL_{\max} = 20 + 5 - 3 + 20 - 3 + 75 - 10 = 104 \text{ dB}$$

Identical, so the two links are perfectly balanced and the range is the same, 139.8 m. Quote both and say which limits.

- **Flag the inconsistency, it earns credit:** $A_{mu} = 43 \text{ dB}$ is Okumura's value at *tens of km*, not at 140 m. Using it at this range makes the model self-inconsistent. The honest engineering answer is that **with the stated 43 dB clutter allowance the link only closes over about 140 m**, and that either the clutter figure or the antenna gains in the question are unrealistic.

Problem 16 Feasibility of a 10 km link at 2.4 GHz

PIN 1/22 74 Bh 12 · a “prove it does not work” question, so the answer must end with a verdict

QUESTION AS PRINTED 2074 Bhadra · Q3 · [12]

Estimate the feasibility of a 10-km wireless link in suburban area, with one access point and one client radio, using Okumura model for path loss. The median attenuation value is 20 dB and gain due to environment is 13 dB. The height of access point antenna is 100 m and that of client antenna is 10 m:

- Access point is connected to antenna with 5-dBi gain, with a transmitting power of 20-dBm and a receive sensitivity of −80-dBm
- Client is connected to antenna with 20-dBi gain, with a transmitting power of 15-dBm and a receive sensitivity of −75-dBm
- Cables in both systems are short, with a loss of 3-dB at each side at 2.4-GHz frequency of operation.

– Step 1: free space loss at 10 km, 2.4 GHz.

$$\lambda = \frac{3 \times 10^8}{2.4 \times 10^9} = 0.125 \text{ m}$$

$$L_F = 32.44 + 20 \log(2400) + 20 \log(10) = 32.44 + 67.60 + 20 = 120.05 \text{ dB}$$

– Step 2: Okumura median path loss.

$$G(h_{te}) = 20 \log(100/200) = -6.02 \text{ dB}, \quad G(h_{re}) = 20 \log(10/3) = +10.46 \text{ dB}$$

$$L_{50} = 120.05 + 20 - (-6.02) - 10.46 - 13 = \boxed{122.61 \text{ dB}}$$

– Step 3: forward link, AP → client.

$$P_r = 20 + 5 - 3 + 20 - 3 - 122.61 = -83.61 \text{ dBm}$$

Client sensitivity is −75 dBm, so the margin is

$$-83.61 - (-75) = \boxed{-8.61 \text{ dB}} \quad (\text{shortfall})$$

– Step 4: reverse link, client → AP.

$$P_r = 15 + 20 - 3 + 5 - 3 - 122.61 = -88.61 \text{ dBm}$$

AP sensitivity is −80 dBm, so the margin is

$$-88.61 - (-80) = \boxed{-8.61 \text{ dB}} \quad (\text{shortfall})$$

– Step 5: verdict. Both directions fall 8.6 dB short, so the 10 km link is NOT feasible as specified, and it fails symmetrically, so no single end can be blamed.

– Step 6: what distance would work? Set the margin to zero:

$$L_{50, \max} = 122.61 - 8.61 = 114.00 \text{ dB}$$

$$L_F = 114.00 - 20 + (-6.02) + 10.46 + 13 = 111.44 \text{ dB}$$

$$d = \frac{0.125}{12.566} 10^{111.44/20} = 0.009947 \times 3.729 \times 10^5 = \boxed{3.71 \text{ km}}$$

– How to fix it (worth stating): raise the antenna gains, raise the AP height (each doubling of h_{te} buys 6 dB), or accept a shorter link. Raising transmit power by 8.6 dB would also do it but may breach the licence-exempt limit at 2.4 GHz.

Problem 17 Power delay profile: delay spread and coherence bandwidth

HOT 5/22 70 Ma 4+4, 71 Ma 6, 75 Bh 8, 81 Bh 6 · four different papers, one method

The method, once

– Step 1: convert every dB power to a linear ratio, $P_k = 10^{(\text{dB})/10}$. Never average in dB.

– Step 2: mean excess delay $\bar{\tau} = \frac{\sum P_k \tau_k}{\sum P_k}$

– Step 3: second moment $\overline{\tau^2} = \frac{\sum P_k \tau_k^2}{\sum P_k}$

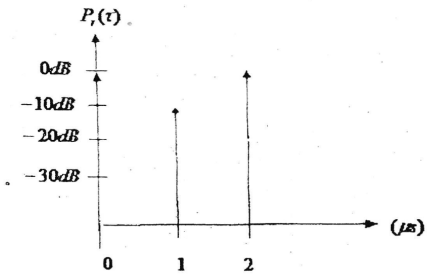
- **Step 4: rms delay spread** $\sigma_\tau = \sqrt{\tau^2 - \bar{\tau}^2}$
- **Step 5: coherence bandwidth** $B_c = \frac{1}{50\sigma_\tau}$ (90 %) or $\frac{1}{5\sigma_\tau}$ (50 %).

17.1 The 70 Ma profile (4+4)

QUESTION AS PRINTED 2070 Magh · Q4 · [4+4]

Define Doppler spread. Describe the types of small scale fading based on Doppler spread. Calculate the mean excess delay and rms delay spread for the multipath profile given below. Estimate the 90% and 50% coherence bandwidth of the channel.

Read the stem plot on the right carefully: it carries **three** components, 0 dB at $\tau = 0$, -10 dB at $\tau = 1 \mu\text{s}$ and 0 dB at $\tau = 2 \mu\text{s}$. **The $\tau = 0$ ray is the second arrowhead sitting on the 0 dB tick, drawn over the vertical axis. Miss it and every answer below is wrong.**



The profile as printed on the 70 Ma paper. **Two arrowheads on the vertical line: upper one is the axis, the one at 0 dB is the $\tau = 0$ ray.**

$\tau_k (\mu\text{s})$	dB	P_k	$P_k \tau_k$	$P_k \tau_k^2$
0	0	1.0	0.0	0.0
1	-10	0.1	0.1	0.1
2	0	1.0	2.0	4.0
sum		2.1	2.1	4.1

$$\bar{\tau} = \frac{2.1}{2.1} = 1.000 \mu\text{s}$$

$$\overline{\tau^2} = \frac{4.1}{2.1} = 1.952 \mu\text{s}^2$$

$$\sigma_\tau = \sqrt{1.952 - (1.000)^2} = \sqrt{0.952} = 0.976 \mu\text{s}$$

$$B_c(90\%) = \frac{1}{50 \times 0.976 \times 10^{-6}} = 20.5 \text{ kHz}$$

$$B_c(50\%) = \frac{1}{5 \times 0.976 \times 10^{-6}} = 204.9 \text{ kHz}$$

- **Maximum excess delay** within 10 dB of the peak: peak is 0 dB, and the 1 μs ray is exactly 10 dB down, so all three count and $\tau_{10\text{dB}} = 2 - 0 = 2 \mu\text{s}$.
- **Sanity check:** $\bar{\tau}$ falls on exactly 1 μs because the two equal 0 dB rays at 0 and 2 μs balance about the centre. A clean number is the sign you read the plot right.

17.2 The 75 Bh profile: greatest symbol rate without an equalizer (8)

QUESTION AS PRINTED 2075 Bhadra · Q5 · [8]

Determine the smallest symbol period T_s , and thus the greatest symbol rate that must be sent through RF channel with given power delay profile without using an equalizer.

Power [dB]	0	0	-10	-20
Delay [us]	0	50	75	100

Modulation provides suitable BER performance whenever $\sigma_\tau / T_s \leq 0.1$.

$\tau_k (\mu\text{s})$	dB	P_k	$P_k \tau_k$	$P_k \tau_k^2$
0	0	1.00	0.0	0.0
50	0	1.00	50.0	2500.0
75	-10	0.10	7.5	562.5
100	-20	0.01	1.0	100.0
sum		2.11	58.5	3162.5

$$\bar{\tau} = \frac{58.5}{2.11} = 27.725 \mu\text{s}$$

$$\overline{\tau^2} = \frac{3162.5}{2.11} = 1498.82 \mu\text{s}^2$$

$$\sigma_\tau = \sqrt{1498.82 - (27.725)^2} = \sqrt{1498.82 - 768.68} = \sqrt{730.14} = \boxed{27.02 \mu\text{s}}$$

- Apply the BER condition.

$$\frac{\sigma_\tau}{T_s} \leq 0.1 \Rightarrow T_s \geq \frac{\sigma_\tau}{0.1} = 10 \sigma_\tau$$

$$T_s \geq 10 \times 27.02 \mu\text{s} = \boxed{270.2 \mu\text{s}}$$

- Greatest symbol rate.

$$R_s = \frac{1}{T_s} = \frac{1}{270.2 \times 10^{-6}} = \boxed{3.70 \text{ ksymbols/s}}$$

- Interpretation:** this channel is so dispersive that without an equalizer the link is limited to under 4 ksym/s. That single number is the practical argument for equalization in Ch 5.

17.3 The 71 Ma profile: is the channel flat for AMPS and for GSM? (6)

QUESTION AS PRINTED 2071 Magh · Q3 (b) · [6]

A wireless channel is characterized by the following power-delay profile:

Power [dB]	0	-10	-20	-23
Delays [ns]	0	100	200	400

Determine the root mean square (rms) delay spread and the 90% coherence bandwidth of the above channel. Is this channel flat fading or frequency selective fading for:

- An AMPS system with transmission bandwidth 30 kHz?
- A GSM system with transmission bandwidth 200 kHz?

τ_k (ns)	dB	P_k	$P_k \tau_k$	$P_k \tau_k^2$
0	0	1.0000	0.00	0
100	-10	0.1000	10.00	1000
200	-20	0.0100	2.00	400
400	-23	0.0050	2.00	798
sum		1.1150	14.005	2201.6

$$\bar{\tau} = \frac{14.005}{1.1150} = 12.560 \text{ ns}$$

$$\overline{\tau^2} = \frac{2201.6}{1.1150} = 1974.78 \text{ ns}^2$$

$$\sigma_\tau = \sqrt{1974.78 - (12.560)^2} = \sqrt{1974.78 - 157.76} = \sqrt{1817.02} = \boxed{42.63 \text{ ns}}$$

$$B_c(90\%) = \frac{1}{50\sigma_\tau} = \frac{1}{50 \times 42.63 \times 10^{-9}} = \boxed{469.2 \text{ kHz}}$$

- (i) **AMPS, 30 kHz.** 30 kHz $\ll$ 469.2 kHz, so $B_s \ll B_c \rightarrow$ **FLAT fading**. No equalizer needed.
- (ii) **GSM, 200 kHz.** 200 kHz < 469.2 kHz, so strictly the channel is still flat, but only by a factor of 2.3. **That is not "much less than", so GSM is marginal on this channel** and a real GSM receiver would still run its equalizer. State it exactly this way: strictly flat, practically marginal.
- If the 50 % definition is used instead, $B_c = 1/(5\sigma_\tau) = 4.69 \text{ MHz}$ and both systems are comfortably flat. **Always say which definition you used.**

17.4 The 81 Bh profile (6)

QUESTION AS PRINTED 2081 Bhadra · Q5 · [4+6]

Compare Okumura model with Hata model. Calculate mean excess delay, rms delay spread and maximum excess delay for the multipath profile $P_r(\tau)$ given as: $P_r(\tau) = (-20\text{dB}, -10\text{dB}, -10\text{dB}, 0\text{dB})$ for $\tau = (0, 1, 2, 5)$ second. Also estimate 50% coherence bandwidth.

"second" is the paper's own slip: the delays are microseconds. Work in μs and say so.

τ_k	dB	P_k	$P_k \tau_k$	$P_k \tau_k^2$
0	-20	0.01	0.00	0.00
1	-10	0.10	0.10	0.10
2	-10	0.10	0.20	0.40
5	0	1.00	5.00	25.00
sum		1.21	5.30	25.50

$$\bar{\tau} = \frac{5.30}{1.21} = \boxed{4.380 \text{ s}}$$

$$\overline{\tau^2} = \frac{25.50}{1.21} = 21.074 \text{ s}^2$$

$$\sigma_\tau = \sqrt{21.074 - (4.380)^2} = \sqrt{21.074 - 19.183} = \sqrt{1.891} = \boxed{1.375 \text{ s}}$$

- **Maximum excess delay (10 dB).** The peak is 0 dB at $\tau = 5$. Components within 10 dB of it are those at ≥ -10 dB, ie $\tau = 1, 2, 5$. So

$$\tau_{10\text{dB}} = 5 - 1 = \boxed{4 \text{ s}}$$

$$B_c(50\%) = \frac{1}{5 \times 1.375} = \boxed{0.1455 \text{ Hz}}$$

- **Source defect, state it:** the paper prints the delays in **seconds**. A real multipath channel has delays of **microseconds**, and a 0.15 Hz coherence bandwidth is physically impossible. **Solve it as printed**, then add one line noting that with μs the answers become $\bar{\tau} = 4.38 \mu\text{s}$, $\sigma_\tau = 1.375 \mu\text{s}$, $B_c(50\%) = 145.5 \text{ kHz}$.

Problem 18 Doppler shift and coherence time

HOT 4/22 78 Ch, 81 Ba, 80 Ch, 82 Bh

PROBLEM not a PYQ · the standard Rappaport example verified/added

A US digital cellular system operating at $f_c = 900 \text{ MHz}$ serves a vehicle moving at 70 km/h . Compute the received carrier frequency when the mobile moves (a) directly towards the transmitter, (b) directly away from it, and (c) perpendicular to the direction of arrival. Hence find the Doppler spread and the 50% coherence time.

Four papers ask for the Doppler *derivation* in prose (78 Ch, 81 Ba, 80 Ch, 82 Bh) and none prints numbers. This worked example is what turns that derivation into a full answer. verified/added

- **Step 1: convert the velocity. Always to m/s.**

$$v = 70 \frac{\text{km}}{\text{h}} \times \frac{1000}{3600} = 19.444 \text{ m/s}$$

- **Step 2: wavelength and maximum Doppler shift.**

$$\lambda = \frac{3 \times 10^8}{900 \times 10^6} = 0.3333 \text{ m}$$

$$f_m = \frac{v}{\lambda} = \frac{19.444}{0.3333} = \boxed{58.33 \text{ Hz}}$$

- **(a) Towards the transmitter, $\theta = 0^\circ$, $\cos \theta = +1$:**

$$f_d = +58.33 \text{ Hz} \Rightarrow f = 900 \text{ MHz} + 58.33 \text{ Hz} = \boxed{900.00005833 \text{ MHz}}$$

- **(b) Away, $\theta = 180^\circ$, $\cos \theta = -1$:**

$$f_d = -58.33 \text{ Hz} \Rightarrow f = \boxed{899.99994167 \text{ MHz}}$$

- (c) **Perpendicular**, $\theta = 90^\circ$, $\cos \theta = 0$:

$$f_d = 0 \Rightarrow f = \boxed{900 \text{ MHz exactly}}$$

- **Step 3: coherence time.**

$$T_c \approx \frac{0.423}{f_m} = \frac{0.423}{58.33} = 7.25 \text{ ms}$$

- **Interpretation:** the channel stays essentially constant for only 7.25 ms. A GSM burst is 0.577 ms, so the channel is safely static within one burst, which is exactly the design intent. A data frame lasting longer than 7 ms would see the channel change under it.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELTP	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise

Short	Means	Short	Means
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
ASK	amplitude shift keying	FSK	frequency shift keying
BPSK	binary phase shift keying	QPSK	quadrature phase shift keying
OQPSK	offset (staggered) QPSK	DPSK,	differential (binary) PSK
		DBPSK	
CPFSK	continuous phase FSK	MFSK,	M-ary FSK, M-ary PSK
		MPSK	
DSSS	direct sequence spread spectrum	FHSS	frequency hopped spread spectrum
SS, SSM	spread spectrum (modulation)	PG	processing gain
PAPR	peak-to-average power ratio	CP	cyclic prefix
FFT, IFFT	(inverse) fast Fourier transform	ICI	inter-carrier interference
PSD	power spectral density	CFO, SFO	carrier, sampling frequency offset
NRZ, RZ	non-return, return to zero	AMI	alternate mark inversion
AM, FM, PM	amplitude, frequency, phase modulation	DSB, SSB, VSB	double, single, vestigial sideband
S/P, P/S	serial-parallel converter	D/A, A/D	digital-analog converter
LPF, BPF	low pass, band pass filter	ISM	industrial, scientific, medical band
SC-FDMA	single carrier FDMA	<i>BT</i>	GMSK filter bandwidth × symbol time
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
SSMA	spread spectrum multiple access	SDMA	space division multiple access
FHMA	frequency hopped multiple access	DSMA	direct sequence multiple access (= CDMA)
FCDMA	hybrid FDMA/CDMA	DS/FHMA	hybrid direct sequence / frequency hopped
TCDMA	time division CDMA	TDFH	time division frequency hopping
IM	intermodulation (product)	MU-MIMO	multi-user MIMO
WLAN	wireless local area network	WMAN	wireless metropolitan area network
WPAN	wireless personal area network	AP	access point
BSS, IBSS	basic service set, independent (ad hoc) BSS	ESS	extended service set
CSMA/CA	carrier sense multiple access with collision avoidance	CSMA/CD	... with collision detection
DCF	distributed coordination function	PCF	point coordination function
RTS/CTS	request to send / clear to send	CCK	complementary code keying
WEP, WPA	wired equivalent privacy, WiFi protected access	U-NII	unlicensed national information infrastructure band
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
PCN	personal communication network	SMS	short message service
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet

data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

One clash to watch: **FCC** is the *forward control channel* throughout these notes. The US regulator is always written out as the Federal Communications Commission.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.